Question:
A merchant wants to make a choice of purchase between 2 purchase plans: jewelry worth $5,000 or electronic gadgets worth $8,000. His financial advisor speculates that the jewelry market will go up 2.5% while the electronic gadgets market will rise 1.2% within the same month. If the merchant is looking to maximize profit at the end of this month by making a choice, how much profit would this be?

Reasoning + Answer:

Let's break down the problem step by step to determine which purchase plan will maximize the profit at the end of the month and by how much.

Step 1: Calculate the profit from the jewelry purchase plan

1. **Initial value of jewelry**: $5,000
2. **Market increase for jewelry**: 2.5%

To find the profit from the jewelry purchase plan, we need to calculate the new value of the jewelry after the market increase and then determine the profit.

Calculation:
- Market increase in dollars:
\[
\text{Increase} = 5000 \times \frac{2.5}{100}
\]
\[
\text{Increase} = 5000 \times 0.025
\]
\[
\text{Increase} = 125
\]

- New value of the jewelry:
\[
\text{New value} = 5000 + 125
\]
\[
\text{New value} = 5125
\]

- Profit from jewelry:
\[
\text{Profit} = \text{New value} - \text{Initial value}
\]
\[
\text{Profit} = 5125 - 5000
\]
\[
\text{Profit} = 125
\]

Step 2: Calculate the profit from the electronic gadgets purchase plan

1. **Initial value of electronic gadgets**: $8,000
2. **Market increase for electronic gadgets**: 1.2%

To find the profit from the electronic gadgets purchase plan, we need to calculate the new value of the gadgets after the market increase and then determine the profit.

Calculation:
- Market increase in dollars:
\[
\text{Increase} = 8000 \times \frac{1.2}{100}
\]
\[
\text{Increase} = 8000 \times 0.012
\]
\[
\text{Increase} = 96
\]

- New value of the electronic gadgets:
\[
\text{New value} = 8000 + 96
\]
\[
\text{New value} = 8096
\]

- Profit from electronic gadgets:
\[
\text{Profit} = \text{New value} - \text{Initial value}
\]
\[
\text{Profit} = 8096 - 8000
\]
\[
\text{Profit} = 96
\]

Step 3: Compare the profits and determine the maximum profit

- Profit from jewelry: $125
- Profit from electronic gadgets: $96

The maximum profit is from the jewelry purchase plan.

Final Answer
<think>
1. Jewelry profit: $125
2. Electronic gadgets profit: $96
3. Maximum profit: $125
</think>
<answer>125</answer>